

Chinese pharmacologist Tu Youyou painstakingly researched thousands of compounds for their antimalarial properties at the end of the 1960s—and finally isolated artemisinin, which became the main malaria treatment worldwide. Her discovery eventually earned her a Nobel Prize in 2015.

MILESTONES

INSPIRING ILLNESS

Contracts tuberculosis as a teenager in 1937, and becomes inspired to research medicines.

EARLY RESEARCH

Studies schistosomiasis as part of her first job in 1955, and gains insights into parasite diseases.

HEADS TEAM

Aged 39, builds a team to find a new antimalarial in 1969, after thousands of compounds fail screening.

ANCIENT KNOWLEDGE

In 1971, reads a 1,600-year-old recipe that helps her find an effective way to extract artemisinin.

Tu Youyou was born in Ningbo on China's east coast. Despite an interruption to her education when she had tuberculosis (TB) for 2 years, she went on to study pharmacology at Peking University, where her tutor introduced her to herbal medicine. In 1955, she joined the newly formed Academy of Traditional Chinese Medicine. Here, she researched *Lobelia chinensis* (Chinese lobelia) as a remedy for schistosomiasis (bilharzia), an infection carried by a parasitic worm that was common in parts of China.

Malaria research

Caused by the single-cell *Plasmodium* parasite and transmitted by female *Anopheles* mosquitoes, malaria was on the rise throughout the world, including in China, despite the World Health Organization's global malaria eradication program (1955–1968). Death rates were rising and new treatments were urgently needed due to increasing resistance of the parasite to existing drugs, such as chloroquine. In 1969, as there had not been any new medicines discovered, the Chinese government appointed Tu as head of the secretive "Project 523" on Hainan Island to source new antimalarial drugs from traditional Chinese medicines. Within 3 months, Tu and her team had gathered more than 2,000

“No doubt, traditional Chinese medicine provides a **rich resource.”**

Tu Youyou, 2015

Tu is the chief scientist at the China Academy of Traditional Chinese Medicine, Beijing. The institution currently serves as the World Health Organization Collaborating Centre for Traditional Medicine.